

# Silent ships do not always encounter more fish: comparison of acoustic backscatter recorded by a noise-reduced and a conventional research vessel

Alex De Robertis, Vidar Hjellvik, Neal J. Williamson, and Christopher D. Wilson

De Robertis, A., Hjellvik, V., Williamson, N. J., and Wilson, C. D. 2008. Silent ships do not always encounter more fish: comparison of acoustic backscatter recorded by a noise-reduced and a conventional research vessel. – ICES Journal of Marine Science, 65: 623–635.

The extent to which fish avoid approaching research vessels is an important source of uncertainty in fisheries surveys. Vessels radiate noise at the frequencies where fish hearing is most sensitive, and noise is thus thought to be the primary stimulus for vessel avoidance. In an effort to minimize vessel avoidance, international standards for noise emission by research vessels have been established. Although vessels meeting these criteria are now in service, the effectiveness of noise quietening on vessel avoidance remains poorly understood. The new, noise-reduced, RV “Oscar Dyson” (OD) will augment the conventionally constructed research vessel, “Miller Freeman” (MF) and serve as the primary platform in conducting acoustic surveys of walleye pollock (*Theragra chalcogramma*) in Alaska. To investigate whether noise-reduction measures result in differential avoidance, which would bias the pollock abundance time-series, we conducted an inter-vessel comparison of acoustic backscatter recorded by OD and MF during a survey of walleye pollock in 2006 in the eastern Bering Sea. Overall, we found no evidence for differences in vessel avoidance that would impact the echo integration results of adult pollock. Analysis of pollock depth distributions from both vessels suggests that there is a comparatively greater diving response to OD, with the reaction taking place primarily after the vessel has passed and for fish shallower than 90 m. Given that the change in vertical distribution is after the fish have been detected by the echosounder, this reaction should not influence echo-integration measurements. The results indicate that use of the OD rather than the MF is unlikely to bias the Bering Sea survey time-series through changes in vessel avoidance by adult walleye pollock.

**Keywords:** acoustics, hearing, noise-reduced vessel, survey, vessel avoidance, walleye pollock.

Received 14 May 2007; accepted 27 December 2007; advance access publication 11 March 2008.

A. De Robertis, N. J. Williamson, and C. D. Wilson: National Marine Fisheries Service, Alaska Fisheries Science Center, Resource Assessment and Conservation Engineering Division, 7600 Sand Point Way NE, Seattle, WA 98115, USA. V. Hjellvik: Institute of Marine Research, PO Box 1870, Nordnes N-5817, Bergen, Norway. Present address: Norwegian Institute of Public Health, PO Box 4404, Nydalen, N-0403 Oslo, Norway. Correspondence to A. De Robertis: tel: +1 206 526 4789; fax: +1 206 526 6723; e-mail: alex.derobertis@noaa.gov

## Introduction

Reports in the literature suggest that, under some circumstances, fish detect, react to, and avoid approaching vessels, leading to concern that vessel avoidance will bias fisheries stock-assessment surveys. A body of work (reviewed in Olsen, 1990; Mitson, 1995; Misund, 1997; Fréon and Misund, 1999; Mitson and Knudsen, 2003) has demonstrated that fish can respond to approaching vessels, often by diving towards the seafloor or by moving horizontally out of the vessel's path, with reactions often initiated well before the vessel passes over the fish. Avoidance behaviour appears to be variable, with the same species exhibiting strong avoidance reactions to vessels in some cases and no response in others (Misund, 1997; Fréon and Misund, 1999; Vabø *et al.*, 2002; Skaret *et al.*, 2005). For fish to avoid a vessel at a distance, the stimulus triggering avoidance must propagate away from the vessel so that it is detected by the fish before the vessel arrives. Given that sound propagates a long distance in water, and that fish are generally most sensitive to low-frequency sounds in the range of 10–500 Hz, which coincides with the frequency range

at which the underwater sound radiated from ships is most intense, the stimulus for this avoidance behaviour is thought to be auditory (Engås *et al.*, 1995; Mitson, 1995; Popper, 2003).

Vessel avoidance has the potential to introduce uncertainty and bias in abundance indices, and remains one of the most poorly understood components of error in many fisheries-independent surveys (Misund, 1997; Fréon and Misund, 1999). It is a particular concern for acoustic surveys, where fish abundance is measured in a narrow (typically  $\sim 7\text{--}12^\circ$ ) downward-looking beam, and fish can travel outside of the beam by making small horizontal motions. If fish dive in response to a vessel, they can also enter near-bottom strata where they cannot be detected by echosounders (Ona and Mitson, 1996). Additionally, the acoustic reflectivity of fish at frequencies used for acoustic surveys depends strongly on their orientation relative to the echosounder beam (Foote, 1985; Hazen and Horne, 2004), and diving responses can change the backscatter from fish detected in the beam (Aglen, 1994). Therefore, even modest behavioural responses to an approaching vessel can have a sizable impact on acoustic-survey results.

The concern that vessel noise induces fish avoidance led to the formulation of recommendations for maximum underwater-radiated noise levels produced by research vessels by the International Council for the Exploration of the Seas (ICES; Mitson, 1995). These recommendations include guidelines for low-frequency noise limits to minimize fish avoidance reactions, and also higher frequency limits intended to maximize the performance of acoustic instruments used during fisheries surveys. The specific specifications related to radiated noise were made based on the hearing capabilities of Atlantic herring (*Clupea harengus*) and Atlantic cod (*Gadus morhua*), which have sensitive hearing, and the recommendation is therefore expected to minimize noise-induced vessel avoidance for other species as well. The recommended vessel-noise limit is such that most species are expected not to react to sound produced by the vessel at distances >20 m. The ICES report presented a method to compute acceptable levels of vessel noise based on the assumption that fish avoid the vessel when noise levels exceed their hearing threshold by 30 dB. This reasoning can be applied for any species where hearing thresholds are known. In practice, the recommendations for cod and herring at a vessel speed of 11 knots, hereafter referred to as the ICES recommendation, have been used as a contractual benchmark the specification and construction of new research vessels.

Several vessels have been constructed to comply with the ICES radiated-noise limits. Specialized vessel designs, including quiet hull designs, diesel-electric propulsion, and fixed-pitch propellers, have resulted in substantial reductions in noise levels over a wide frequency range (10–50 kHz), compared with previous research vessels (Mitson and Knudsen, 2003). Although several nations have invested in vessels conforming to ICES recommendations for radiated noise, there have been few studies evaluating the effectiveness of noise-reduced vessels in terms of reducing fish avoidance. A comparison of echo-integration measurements from a noise-reduced vessel and a quiet, autonomous, underwater vehicle indicated similar herring biomass (Fernandes *et al.*, 2000). This observation was interpreted as a lack of fish avoidance to the vessel, but it is unclear whether the herring would have avoided a noisier conventional vessel, making it difficult to evaluate whether noise-reduction altered vessel avoidance in this case. A recent study (Ona *et al.*, 2007) comparing avoidance reactions of over-wintering herring to a conventional research vessel with those to a larger, noise-reduced research vessel revealed that although echosounders aboard both vessels detected similar backscatter, the herring performed stronger diving responses when approached by the noise-reduced vessel. This indicates that the stimuli causing vessel avoidance, and therefore the efficacy of noise-reduction in minimizing avoidance responses under survey conditions remains poorly understood.

The US National Oceanic and Atmospheric Administration (NOAA) is currently building a class of four noise-reduced, fisheries-research vessels intended to conform to the ICES specifications for underwater-radiated noise (Bahtiaran, 2005). The first of these, the RV “Oscar Dyson” (OD), is operating in the North Pacific, where it is scheduled to continue a long time-series of acoustic-abundance surveys of walleye pollock (*Theragra chalcogramma*; Wespestad and Megrey, 1990; Honkalehto *et al.*, 2002) traditionally conducted by the conventional RV “Miller Freeman” (MF). These surveys are used to manage a sizeable fishery in Alaska, particularly in the eastern Bering Sea (Bailey *et al.*, 1999). The extent to which pollock respond to approaching

survey vessels remains unclear. Observations from a buoy-mounted echosounder indicate that avoidance reactions occur in some but not all instances (C. Wilson, unpublished data). Moreover, there is an indication that pollock avoid vessels engaged in trawling more than they do free-running vessels (De Robertis and Wilson, 2006). The “Oscar Dyson” was designed to minimize low-frequency, underwater-noise emission to reduce vessel-avoidance, whereas the “Miller Freeman” was not, so there is concern that biomass indices derived from the two vessels will not be equivalent. To ensure consistent survey results from the two vessels, we undertook an inter-vessel comparison designed to establish if pollock avoid the two ships to a different extent.

## Material and methods

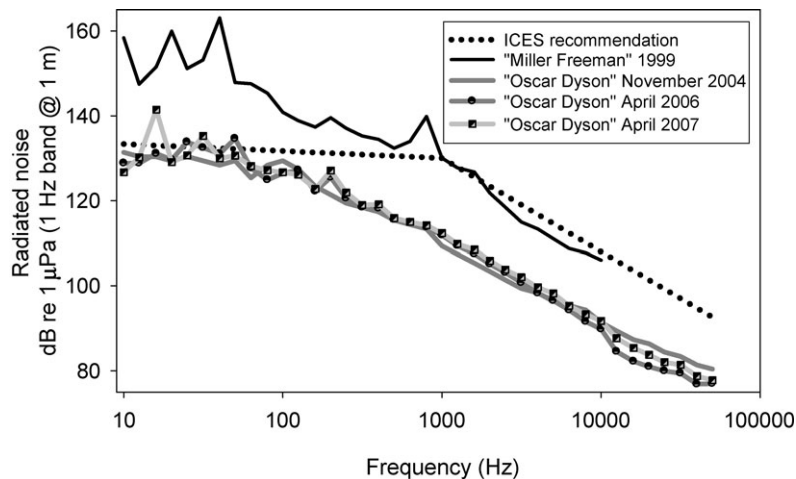
### Research vessels

The NOAA research ships “Oscar Dyson” and “Miller Freeman” are stern trawlers built for fisheries research. Although they are of similar length, the RV “Oscar Dyson” has ~30% more displacement and ~40% more horsepower (Table 1). This vessel was designed to meet the ICES noise-specification maxima and is equipped with noise-control measures such as diesel-electric propulsion, a large fixed-pitch propeller, and sound-dampening material (Bahtiaran, 2005). The RV “Miller Freeman”, on the other hand, is conventionally built with geared diesel propulsion, but was retrofitted with a new propeller, which reduced the radiated noise signature, particularly at high frequencies (Gonzalez *et al.*, 1999). Noise measurements at US Navy installations indicated that auxiliary machinery made minor contributions to radiated noise, and that the propulsion plant and propeller were its primary sources on both vessels (Gonzalez *et al.*, 1999; Thomas and Bradley, 2005; Harmina, 2007).

The RV “Miller Freeman” exceeds the ICES noise-specification maxima at frequencies <2000 Hz which are thought to cause avoidance reactions by fish (Mitson, 1995), but it meets the ICES recommendation at the higher frequencies specified to maximize echosounder performance (Figure 1). OD met the ICES recommendation for radiated noise when it was delivered in 2004, but exceeded the recommendation at several low-frequency bands when re-measured in 2006 and 2007 before and after the experiment (Figure 1). The variability in these repeated measurements

**Table 1.** General specifications of NOAA ships “Miller Freeman” and “Oscar Dyson”.

| Parameter                                    | RV “Miller Freeman”       | RV “Oscar Dyson”   |
|----------------------------------------------|---------------------------|--------------------|
| Year launched                                | 1967                      | 2003               |
| Length (m)                                   | 65.5                      | 63.8               |
| Breadth (m)                                  | 12.8                      | 15.0               |
| Maximum draft with centreboard retracted (m) | 6.4                       | 6.0                |
| Full load displacement (t)                   | 1 920                     | 2 479              |
| Main propulsion                              | Geared diesel             | Diesel electric    |
| Horsepower                                   | 2 150                     | 3 084              |
| Propeller arrangement                        | Single controllable pitch | Single fixed pitch |
| Propeller blades                             | 4                         | 5                  |
| Propeller diameter (m)                       | 3.1                       | 4.3                |



**Figure 1.** Underwater radiated-noise signature of the NOAA RVs “Miller Freeman” and “Oscar Dyson” at speeds of 11 knots plotted along with the ICES recommendation for maxima for RV-radiated noise. The sound-pressure levels have been converted from the one-third octave band to 1 Hz bandwidth measurements following the practice of Mitson (1995). This calculation assumes that the power distribution is uniform within each band. Narrowband tones within a one-third octave band will exceed that shown.

highlights the difficulty of maintaining the noise signature of noise-reduced research vessels over time. After the 2006 tests, but before the vessel comparison, degrading vibration isolation mounts on the diesel generators on OD were identified and replaced because they provided a path for noise to travel to the hull and to radiate into the water. However, when radiated noise was measured again in April 2007, the 16 Hz one-third octave band level exceeded the ICES specification by  $\sim 8$  dB, and the 31.5 Hz band exceeded the ICES recommendation by  $\sim 3$  dB. Further testing indicates that the 16 Hz level is attributable to propeller-blade rate, whereas the 31.5 Hz level is due to the propulsion generators (Edward Bradley, US Navy, pers. comm.). Our experiment was conducted between the 2006 and 2007 measurements, and the precise radiated-noise level at the time of the experiment is unclear. Given the temporal variability in radiated noise produced by OD, and the fact that the radiated noise produced by MF has likely changed since noise measurements were made in 1999, these noise signatures (Figure 1) should be viewed as approximate. Despite the uncertainty in the exact noise signature of the vessels during the comparison, it is reasonable to conclude that OD is much quieter than MF over a broad frequency range.

The method in Mitson (1995), parameterized with laboratory measurements of the hearing sensitivity of walleye pollock (A. Popper and D. Mann, pers. comm.), predicts that pollock will avoid MF at a range of  $\sim 75$  m and OD at a range of 20 m in response to radiated noise levels at  $\sim 100$  Hz when the vessels are travelling at 11 knots. Although there are several assumptions inherent in these calculations, the predicted distances illustrate the potential for pollock to exhibit differential avoidance responses to the two vessels according to differences in underwater-radiated noise levels as the vessels approach.

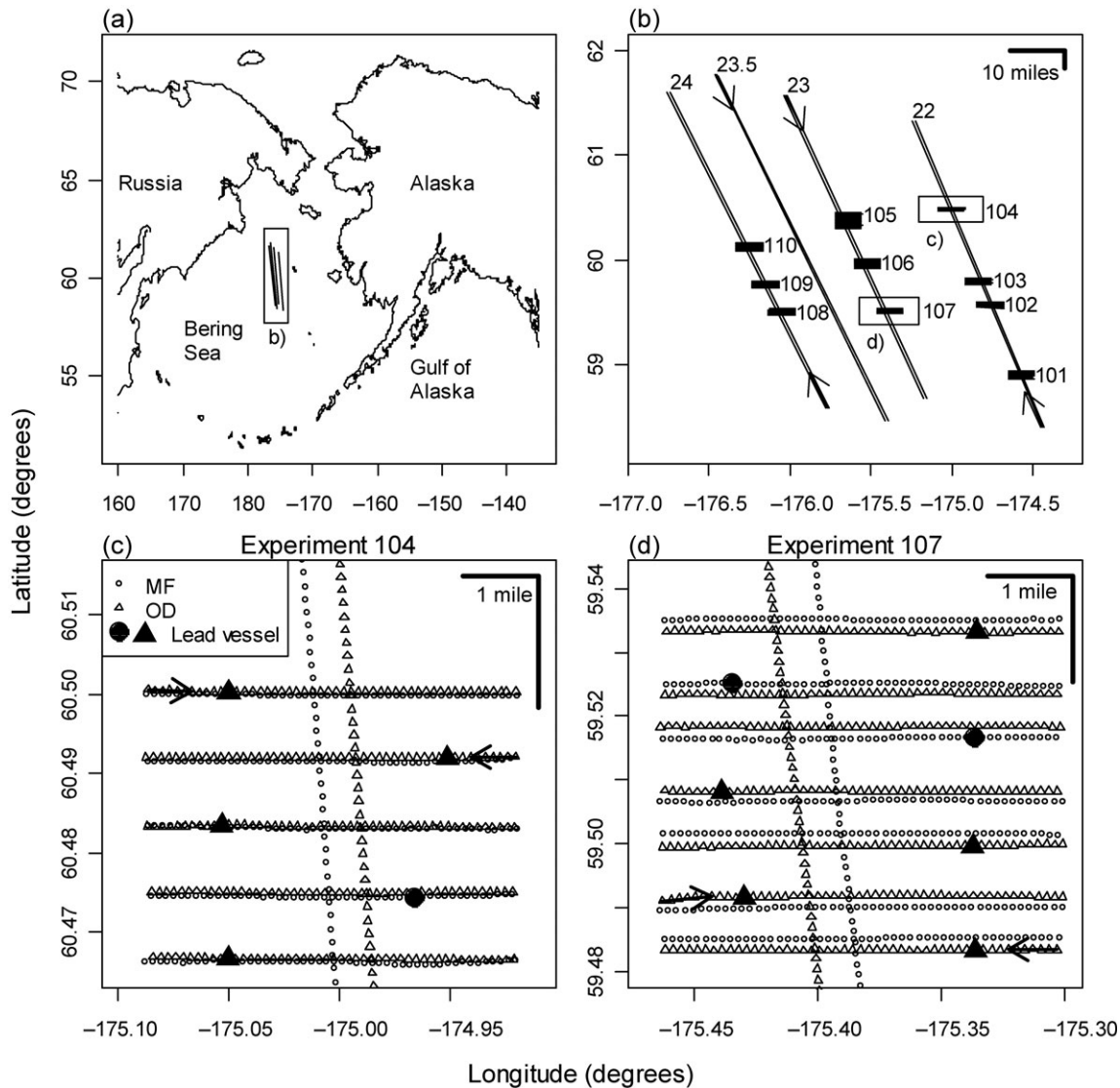
### Study design

An inter-vessel comparison of MF and OD was conducted between 3 and 13 July 2006 concurrent with the National Marine Fisheries Service biennial survey of walleye pollock (Honkalehto *et al.*, 2002) on the eastern Bering Sea shelf conducted by MF (Figure 2a). The vessels surveyed at  $\sim 12$  knots, MF using the standard survey shaft

rpm and propeller-pitch settings used during this survey (185 rpm, 80% pitch), and OD adjusting shaft rpm to match MF's speed. The weather during the experiment was mild, with wind speeds ranging from  $\sim 2$  to 22 knots ( $\bar{x} \pm \text{s.d.}$ ,  $13.9 \pm 4.6$ ), and wave heights  $< 2$  m. The experimental design included a component in which the vessels travelled side by side at a constant separation distance (side-by-side transects), and a component in which the vessels followed each other along short transects (follow-the-leader transects). The side-by-side measurements were made at a separation of 0.5 nautical miles (hereafter referred to simply as miles) along three 170–214-mile transects spaced 20 miles apart as part of the acoustic survey (transects 22–24 in Figure 2). These transects were conducted using standard procedures (Honkalehto *et al.*, 2002), including collecting acoustic data only during daylight. An additional 215-mile transect was added as the vessels returned to port (transect 23.5 in Figure 2b), but only  $\sim 66\%$  of this transect was conducted during daylight because of time constraints. For the side-by-side work, the OD was randomly offset 0.5 miles either east or west of the pre-planned MF survey trackline.

The vessel-separation distance for the side-by-side transects was chosen so that noise from MF was not expected to influence the radiated noise near OD, the quieter vessel. At the closest separation distance of 0.5 miles, noise from the adjacent vessel is subject to  $\sim 60$  dB in propagation loss (assuming spherical spreading), which is substantially more than the difference in radiated noise between the vessels (Figure 1). Therefore, it is unlikely that the presence of the accompanying vessel substantially altered the radiated noise field near either vessel during the experiment.

At 10 locations along the survey trackline, the side-by-side comparison was interrupted to conduct dedicated experiments (101–110; Figure 2b) in which the vessels took turns “following the leader”. The vessels conducted 5.0-mile east- or west-heading transects, with one vessel leading at a distance of  $\sim 1.0$  mile (Figure 2c). Between 5 and 20 transects were conducted during each experiment, a total of 101 transects. Preliminary analysis in the field after the first 36 transects hinted at lower estimates of echo abundance from the trailing vessel, potentially through absorption from bubbles caused by the wake of the leading vessel. As a



**Figure 2.** Experimental design. (a) Location of the experiment, which is shown in detail in (b)–(d). (b) Side-by-side transects (22–24, 23.5) and follow-the-leader experiments (101–110), arrows indicating the direction of travel. (c) Details of experiment 104 overlaid on a portion of transect 22. Each point is the start of a 0.1-mile segment. Arrows on the first two transects indicate the direction of travel. The large symbols indicate the position of the lead vessel when the trailing vessel starts on each transect. (d) Details for experiment 107, where the following vessel was displaced laterally to starboard by 0.1 nautical mile.

consequence of this possible “lead effect”, the trailing vessel was displaced 0.1 mile to the starboard side of the leading vessel in the last 65 transects (experiments 105–110) to avoid the wake of the leading vessel (Figure 2d). Vessel lead was assigned at random for each follow-the-leader transect. These transects were conducted during both daylight and darkness.

The identity of echosign suspected to be pollock was confirmed by targeted fishing with a midwater Aleutian wing trawl (Honkalehto *et al.*, 2002) on both vessels. A trawl haul was made by at least one vessel, and on seven occasions by both vessels, after each follow-the-leader experiment. Additional trawls were conducted at five locations along the side-by-side tracklines. Pollock dominated all catches, accounting for an average of 95.5% of catch by weight, with a range of 78–100%. The pollock were primarily adults, with some 97.5% of pollock in each catch exceeding a fork length of 30 cm (range of 81–100%).

### Acoustic-data collection

Both ships were equipped with 18, 38, 120, and 200 kHz Simrad split-beam EK60 echosounders (reference to trade names does not imply endorsement by the National Marine Fisheries Service, NOAA). The transducers (models ES18, ES38-B, ES120-7C, ES200-7C) were mounted on retractable centreboards at depths of  $\sim 9.1$  m, which reduces bias in echo-integration measurements caused by shallow bubble layers (Dalen and Løvik, 1981; Ona and Traynor, 1990). The Simrad ER60 programme (version 2.1.2) was used to configure and control the echosounders on both vessels. The echosounders were operated at power settings recommended by the manufacturer (Simrad, 2002) to minimize range-dependent losses attributable to harmonic distortion (Tichy *et al.*, 2003). User-selectable parameters such as pulse length (1 ms), ping interval ( $1 \text{ s}^{-1}$ ), sound speed ( $1470 \text{ m s}^{-1}$ ), frequency-dependent absorption coefficients, and



bottom-detection parameters were set to the same values for both vessels. Other acoustic instruments were either turned off or synchronized to the EK60 echosounders to avoid acoustic interference.

The on-axis sensitivity of the echosounders was calibrated using the standard-sphere technique (Foote *et al.*, 1987) on several occasions during the survey. For the MF, calibrations were conducted on four occasions between 4 June and 25 July. For the OD, calibrations were conducted at the beginning and end of the comparison experiment, although two replicate measurements at 38 kHz were conducted each time. To evaluate the impact of a potential “sphere effect” attributable to the use of different standard spheres aboard the two vessels, we calibrated the 38 kHz echosounder aboard the MF with each of the spheres previously used to calibrate either ship. In addition, the “lobes” polynomial calibration method (Simrad, 2003) was performed at each calibration to monitor for changes in beam pattern during the experiment.

The MF's 18 kHz echosounder exhibited noise at short ranges, probably through ringing after the transmit pulse or detection of back-radiation of the transmitted pulse. This problem has been identified as a defect in the transducer, and although it limits short-range (<40 m) data analysis using the integration threshold applied in this study, it does not affect the data analysis at longer ranges. In addition, the 200 kHz system aboard MF appeared to have a range-dependent bias. Compared with other frequencies on the same vessel, as well as the 200-kHz system aboard OD, the echo intensity at 200 kHz decreased substantially with depth (Hjellvik and De Robertis, 2007). Because of these problems, 18 kHz data shallower than 40 m, and all of the 200 kHz data, were excluded from further analysis.

### Acoustic-data processing

Acoustic data were post-processed using Sonardata echoview version 3.50.54. The mean values (averaged in linear units) of integration gain from all available calibrations were applied to each frequency. The manufacturer-supplied estimates of equivalent beam angle (EBA) were adjusted by the square of the ratio of the sound speed observed under nominal field conditions ( $1470 \text{ m s}^{-1}$ ) and during Simrad's transducer calibration to make a first-order correction for the effects of sound speed (cf. Demer, 2004). The backscatter was primarily assigned to two categories: pollock and a near-surface class (mix) whose identity remains poorly characterized but is thought to consist primarily of jellyfish, macrozooplankton, and age-0 pollock. When combined, the pollock and near-surface mix categories accounted for 97.5% of the water-column backscatter (82.5% pollock, 17.0% mix). Pollock occurred in a wide range of echosign types ranging from scattered fish to dense schools, whereas the near-surface mix class appeared as a continuous band in the upper 40 m. Usually, echosign assignment was straightforward owing to the clear vertical separation of these classes. However, at times a portion of the pollock ascended at night into near-surface waters, where the mix was located (Figure 3). Efforts were taken to minimize the impact of subjective judgements on echosign classification: one person inspected and compared all the echograms from the two vessels as they were displayed simultaneously, and the boundary lines separating the two classes in the MF data were overlaid on the OD data, and were used to guide the final assignments for the OD data.

An integration threshold of  $-70 \text{ S}_v$  was applied at 18 and 38 kHz, and a  $-60 \text{ S}_v$  threshold was used at 120 kHz to suppress low-intensity but persistent backscatter attributable to scattering from zooplankton that overlapped with the pollock backscatter. In some areas at the beginning and end of the side-by-side transects where very few pollock were present, the plankton backscatter was observed above the  $-60 \text{ dB}$  integration threshold. These areas,  $\sim 40\%$  of the total trackline for the side-by-side transects, were excluded from further analysis. To ensure consistent spatial coverage across frequencies, we also did not analyse the areas contaminated by plankton backscatter at 18 and 38 kHz.

Bottom depths in the areas retained for analysis averaged 122.5 m and ranged between 101.6 and 136.8 m. Acoustic data from 15 m from the surface to 3 m off bottom were integrated at a 1-m vertical and 0.1-mile horizontal resolution. Data within 3 m of the bottom were excluded so that the results would conform to the data used for stock-assessment purposes (Wespestad and Megrey, 1990). In addition to the water column integrations, echoes from the seafloor were integrated. These bottom integrations were used to compare echosounder performance independently of the on-axis calibration (Johannesson and Mitson, 1983). The bottom-integration zone was defined by extending from the sounder-detected bottom to 25 m below this point.

### Statistical analysis

The analysis was conducted following the approach developed in Kieser *et al.* (1987). The echo-integration measurements were modelled as

$$s_{A,ij} = \alpha_j \rho_i \varepsilon_{ij}, \quad i = 1, \dots, n, \quad j = \text{OD, MF}, \quad (1)$$

where  $s_{A,ij}$  is the nautical-area scattering coefficient ( $s_A$ ,  $\text{m}^2 \text{ mile}^{-2}$ ) recorded at transect segment  $i$  by vessel  $j$ ,  $\rho_i$  the true fish density at transect segment  $i$ ,  $\alpha_j$  a vessel-specific scaling factor, and  $\varepsilon_{ij}$  lognormally-distributed random noise. We were interested in the vessel ratio  $R = \alpha_{\text{OD}}/\alpha_{\text{MF}}$ . Defining

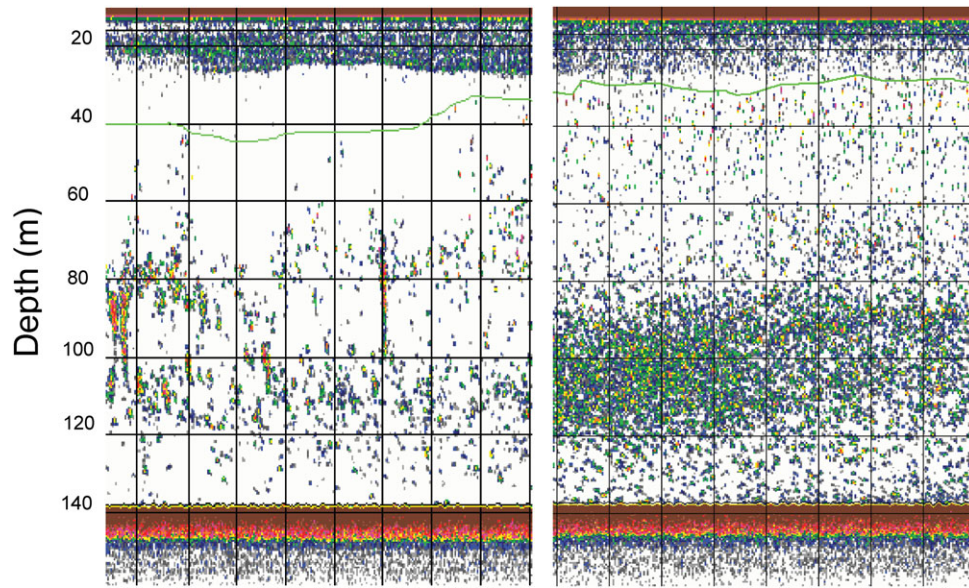
$$d_i = \ln(s_{A,i,\text{OD}}) - \ln(s_{A,i,\text{MF}}) = \ln(\alpha_{\text{OD}}) - \ln(\alpha_{\text{MF}}) + e_i, \quad (2)$$

where  $e_i = \ln(\varepsilon_{i,\text{OD}}) - \ln(\varepsilon_{i,\text{MF}})$  is normally-distributed random noise,

$$\hat{R} = \exp(\bar{d}), \quad (3)$$

where  $\bar{d} = n^{-1} \sum_{i=1}^n d_i$  is an unbiased estimate of  $R$ . Assuming no autocorrelation in  $d_i$ , the 95% confidence interval (CI) for  $R$  is  $\exp(\bar{d} \pm t_{n-1,0.025} s_{\bar{d}} n^{-1/2})$  where  $t_{n-1,0.025}$  is the 2.5% quantile of the  $t$ -distribution with  $n-1$  degrees of freedom.

At a 0.1-mile horizontal resolution, the observed  $d_i$ s were highly autocorrelated, so we aggregated the  $s_A$  in EDSUs (elementary distance sampling units) of five miles. For the side-by-side transects, this was done by defining every 50th 0.1-mile segment as starting points for the OD EDSUs, and taking the 0.1-mile segments for MF that were closest in time to the starting points for the OD EDSUs as starting points for the MF EDSUs. For the follow-the-leader experiments, the EDSUs were defined as the individual five-mile transects. EDSUs with mean  $s_A < 20$  for a given class for one or both vessels were excluded because at very low densities,  $d_i$  can easily become very large or very small if one vessel happens to detect fish that the other vessel does not.



**Figure 3.** Examples of 38 kHz echosign during the day (left) and at night (right). The horizontal green lines separate backscatter assigned to a shallow near-surface species mixture, and the echosign from walleye pollock in the deeper strata. Vertical lines represent integration intervals of 0.1 mile.

Also, at low  $s_A$ , there is less confidence in the species assignment. After excluding the areas containing obvious plankton backscatter at 120 kHz, such low density EDSUs only occurred for the surface mix (47% and 23% of the side-by-side and follow-the-leader EDSUs, respectively) and pollock shallower than 80 m (27% and 24%, respectively).

The ratio  $R$  was estimated for pollock and for the near-surface mix. For pollock, it was also estimated separately for fish situated below or above 80 m deep to test for depth-dependence. We explored the dependence of the vessel ratio on various explanatory variables by examining scatterplots and linearly regressing the  $d_i$  values for the side-by-side transects and the follow-the-leader transects separately on these variables. The variables considered were latitude, longitude, mean pollock depth (see below), pollock-backscatter intensity, and the altitude of the sun above the horizon. For these exploratory analyses, we estimated pollock depth and backscatter intensity by averaging the results from both vessels over each EDSU.

The mean pollock depth for each EDSU was calculated as

$$\bar{D} = \frac{\sum_D D \times s_{A,D}}{\sum_D s_{A,D}}, \quad (4)$$

where  $D = 16, 17, \dots$  is depth (m), and  $s_{A,D}$  is the  $s_A$  in the depth interval from  $D-1$  to  $D$ . The  $p\%$  depth quantiles  $q_p$  ( $p = 10, 20, \dots, 90$ ) of the fish distribution were also calculated by linear interpolation between  $D^+$  and  $D^+ - 1$ , where  $D^+$  is the shallowest depth, such that

$$\sum_{D=16}^{D^+} s_{A,D} > 0.01p \sum_{D=16}^{D_{\max}} s_{A,D}. \quad (5)$$

Depth distributions were computed for all EDSUs. In two of the follow-the-leader transects, one vessel detected a very large

school that the other vessel did not. These transects were omitted from the analysis of the depth distribution.

As an alternative form of echosounder intercalibration, we adjusted the vessel ratios such that the strength of the echo from the bottom during the follow-the-leader experiments would be the same for both vessels. The analysis of the bottom echo is presented in detail in Hjellvik and De Robertis (2007) and so is treated only briefly here. Probably because of changes in the mean angle of incidence between the acoustic beam and the bottom, vessel attitude, particularly vessel list, biased the bottom echo, so we only used five-mile sections where the mean list on both vessels was  $< 1^\circ$  to compute the scaling factor.

The bottom-corrected, vessel-ratio estimate,  $\hat{R}_{\text{corr}}$ , was scaled as follows:

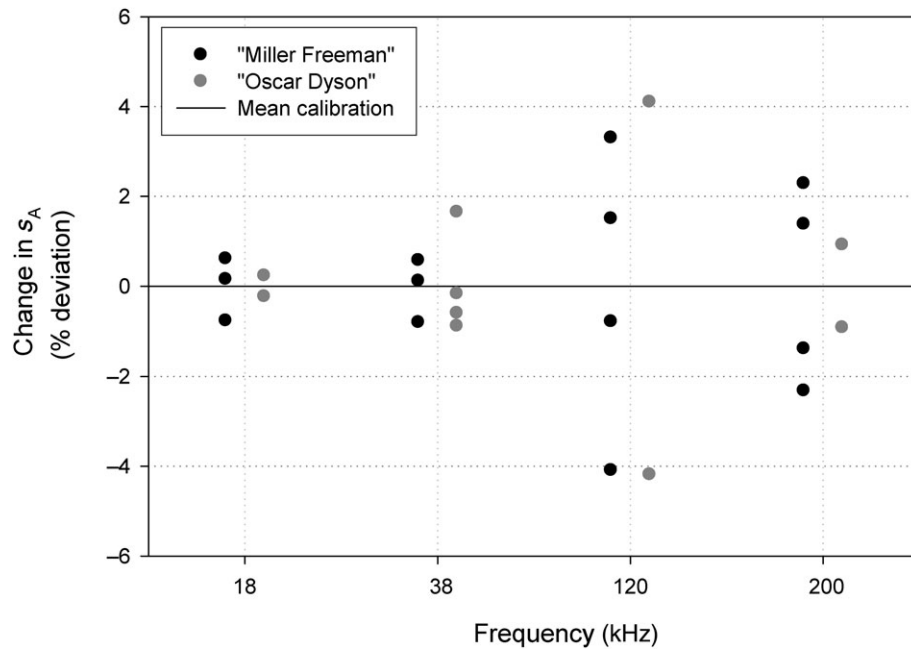
$$\hat{R}_{\text{corr}} = \hat{R}_{\text{obs}} - (\hat{R}_{\text{bot}} - 1), \quad (6)$$

where  $\hat{R}_{\text{obs}}$  is the observed vessel ratio, and  $\hat{R}_{\text{bot}}$  is the vessel ratio computed from the bottom echoes. This procedure is analogous to calibrating the echosounders such that the same frequency instruments will report the same bottom-backscatter strength.

## Results

### Echosounder calibrations

On-axis calibrations were successfully accomplished for both ships except one case for the MF 18 kHz system, during which the appearance of high densities of fish in near-surface waters degraded the calibration results. That calibration was not included in further analyses. The on-axis calibrations exhibited high precision over the study period: if we had chosen to apply any of the individual calibrations rather than the average of all calibrations, we would expect deviations of  $< 1-4\%$  from the observed  $s_A$ , depending on frequency (Figure 4). Repeating the 38 kHz calibration aboard MF with both standard targets indicated that echo integration using either calibration result would lead to differences



**Figure 4.** Results of on-axis calibrations. Each point represents the effect of using a single on-axis sphere calibration on  $s_A$  measurements reported here compared with applying the mean integration gain ( $\overline{\text{Gain}}$ ) from all calibrations combined, as has been applied in this study. Results are expressed as percentage deviation =  $[2(\text{Gain} - \overline{\text{Gain}})/\overline{\text{Gain}}] \times 100$ , where Gain corresponds to the on-axis integration gain in linear units. The calculation accounts for the two-way effects of integration gain on echo integration. Although four calibrations were conducted at 38 kHz for “Miller Freeman”, in two of the cases, results were similar, so just three distinct points are visible.

of  $<2\%$  in echo-integration measurements. This is consistent with the precision observed in repeated measurements on the same sphere (Figure 4), and it indicates that the use of different standard targets aboard the two vessels did not substantially influence calibration results. The beam-pattern estimates produced by the “lobes” procedure did not suggest any changes in beam pattern or echosounder performance during the study period.

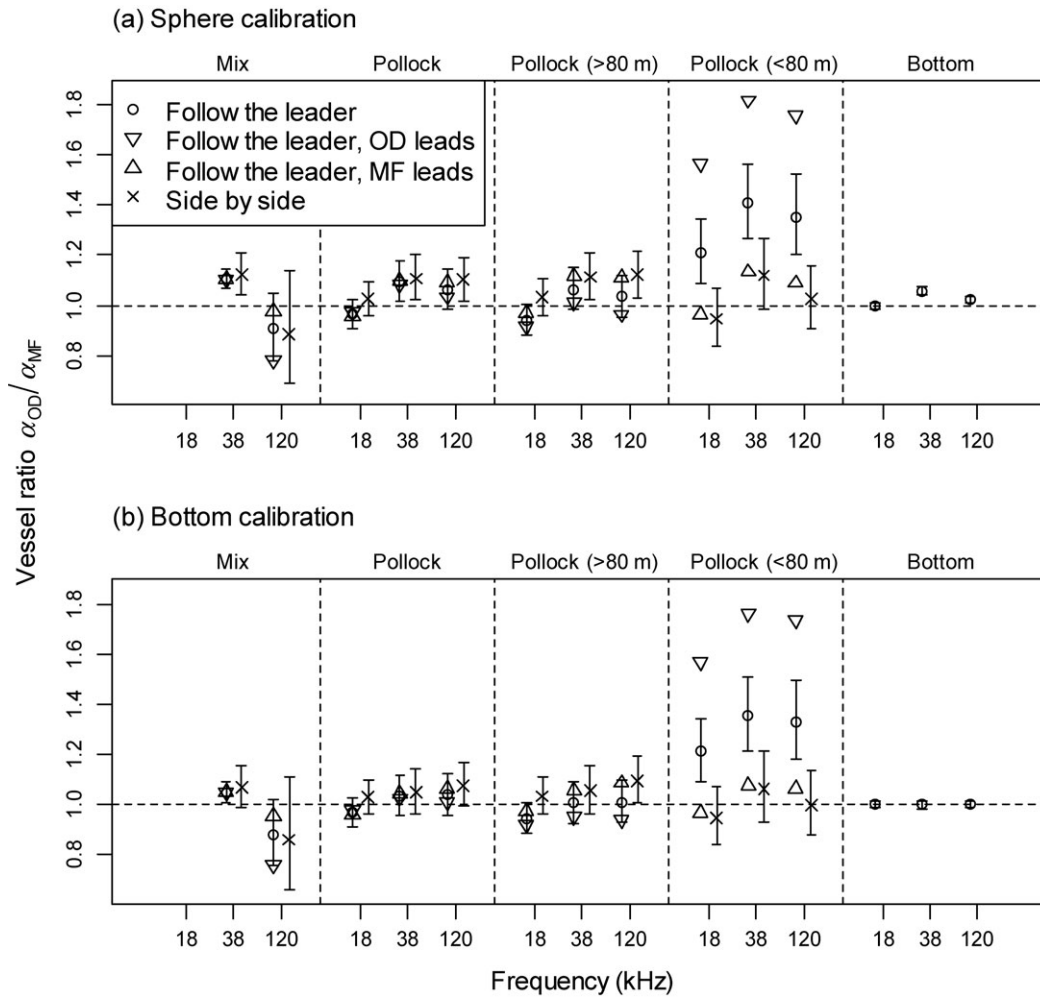
### Vessel-ratio $R$

We evaluated the validity of the two assumptions used in the vessel-ratio analysis: (i)  $d_i$  is not autocorrelated, and (ii)  $\bar{d}$  is  $t$ -distributed. The first lag autocorrelations for  $d_i$  for the EDSUs of the side-by-side transects were  $-0.18$ ,  $-0.16$ ,  $0.21$ , and  $0.44$  for transects 22, 23, 24, and 23.5, respectively. The last number was barely significant at a 5% level, the others were not. For all of the ten follow-the-leader experiments, the first-lag autocorrelation was not significant. Overall, this indicates that the assumption of zero autocorrelation was largely met. Falsely assuming zero autocorrelation would result in CIs that were too narrow. The distribution of  $d_i$  had heavier tails than the normal distribution, and the Shapiro–Wilk normality test applied to  $d_i$  for all transects joined together yielded a  $p$ -value  $< 10^{-9}$ . However, CIs of the mean  $R$  rely on  $\bar{d}$  being  $t$ -distributed rather than  $d_i$  being normally distributed, and according to the central-limit theorem, this may well be the case. To evaluate the validity of this assumption, we applied the  $t$ -test on 100 000 samples of size  $n = 186$  (the number of  $d_i$ s) drawn with replacement from the mean adjusted set  $\{d'_i = d_i - \bar{d}\}$ . At the 5% level, the null hypothesis of  $\bar{d}' = 0$  should then be rejected in 5% of the cases if  $\bar{d}$  is  $t$ -distributed. It was rejected in 5.098% of the cases, so we conclude that the non-normality of the  $d_i$ s is not strong enough to invalidate the CIs.

The suspected “lead effect” that led to the modification of the experimental design where the trailing vessel was displaced laterally by 0.1 mile turned out to be weak. On average, the lead vessel detected slightly more pollock than the trailing vessel before the modification, although this was not significant ( $\bar{d}_i^L = 0.13$ ;  $p > 0.05$ ), but significantly less after the modification ( $\bar{d}_i^L = -0.08$ ;  $p < 0.05$ ). Here,  $d_i^L$  is defined as  $d_i$  in Equation (2), but with OD and MF replaced as lead-vessel and trailing-vessel, respectively. The effect is weak, and both vessels took the lead approximately the same number of times, so a potential lead effect should not have much influence on the overall vessel-ratio estimates from the follow-the-leader experiments.

The estimated vessel ratio was dependent on echosounder frequency for both pollock and the near-surface mix (Figure 5a), with several instances where the 95% CIs excluded one, suggesting significant differences between vessels. However, these differences were frequency-dependent, making them difficult to interpret. For example, MF appeared to detect less backscatter from pollock and near-surface mix at 38 kHz over the entire water column. However, the pattern in the bottom integrations exhibited a similar trend in frequency to that observed for pollock (Figure 5a), indicating that these differences may be due to instrumentation bias rather than differential vessel avoidance by pollock.

When the bottom-ratio adjustment was applied (Figure 5b), the results appeared more consistent across frequencies. The 95% CI for the vessel ratio included one in all six cases examined for pollock and three of four cases examined for the near-surface mix. This suggests that if the bottom adjustment corrected for echosounder differences that were not captured in the on-axis calibrations, both vessels observed equivalent backscatter from pollock and the near-surface mix. No significant relationships



**Figure 5.** Estimated vessel ratios ( $\alpha_{OD}/\alpha_{MF}$ ) with 95% CIs calculated using (a) the sphere calibrations, and (b) correction factors based on the bottom integrations. For the follow-the-leader transects, the vessel ratios stratified by lead vessel are shown as triangles without CIs. The right panel shows the vessel ratio for the bottom echo. Note that the ratios differ from those in Figure 4 of Hjellvik and De Robertis (2007) because the equivalent-beam angles in this study have been corrected for the effects of sound speed during measurement and for an error in which equivalent-beam angles measured on transducers different from those installed on the vessel were accidentally supplied by the manufacturer.

between vessel ratio and latitude, longitude, fish depth, pollock backscatter intensity, or altitude of the sun were detected. The regression  $r^2$  was  $<0.03$  for all comparisons in both the side-by-side and the follow-the-leader transects, and the regression slopes were not significantly different from zero ( $p > 0.1$  in all cases).

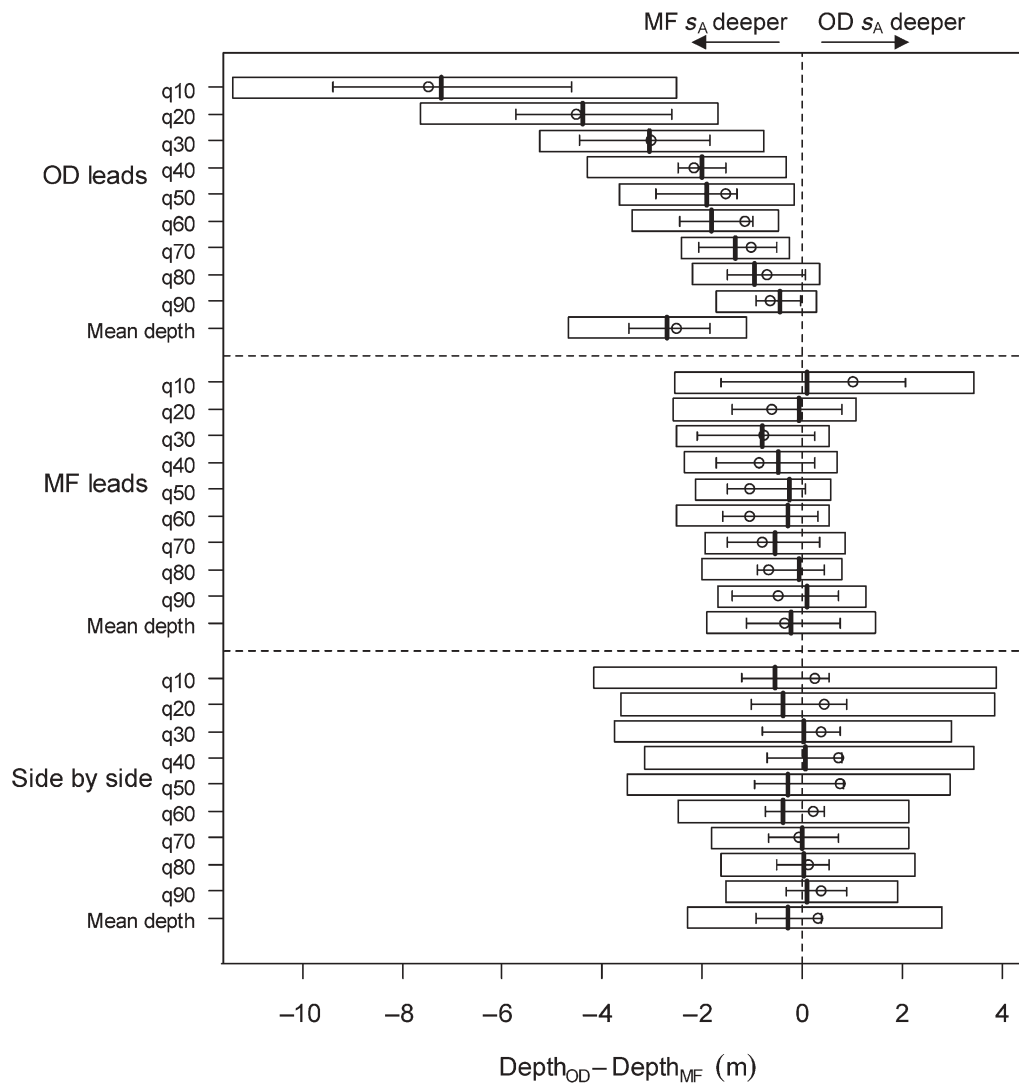
For the near-surface mix and total pollock, there was little evidence that the observations depended on which vessel was in the lead (Figure 5). In contrast, for depth-stratified pollock, there was a strong “lead effect” at all frequencies. At 38 kHz when OD was leading, the vessel ratio  $\hat{R}$  was  $\sim 1.7$  at depths  $<80$  m and close to 1.0 at depths  $>80$  m, whereas when MF was leading  $\hat{R}$  was close to 1.0 in both cases (Figure 5b). This shift in  $\hat{R}$  from strongly positive in the upper water column to slightly negative at greater depths when OD led may indicate a displacement of fish from the upper to the lower water column, this change occurring between the time OD’s transducer measured the fish and the time MF arrived  $\sim 5$  min later. To further address this hypothesis, we examined the vertical distribution of backscatter.

### Vertical distribution

The vertical distribution of pollock echosign was similar for both vessels during the side-by-side transects and the follow-the-leader transects when MF led (Figure 6). The difference in mean pollock-backscatter depth observed by the two vessels (depth<sub>OD</sub> – depth<sub>MF</sub>; mean  $\pm$  95% CI) was  $0.20 \pm 0.80$  m for the side-by-side and  $-0.34 \pm 0.72$  m for the MF-led transects, respectively. In contrast, when OD led, pollock backscatter observed by MF’s echosounder was deeper ( $-2.51 \pm 0.88$  m). This change in vertical distribution was not easily discernible when examining echograms from both vessels, because of the relatively small depth change and the temporal variability in the reaction. The depth difference was close to zero in all three cases for the surface mix (data not shown).

The depth difference when OD led was more pronounced for pollock located close to the surface. The depths for lower (shallower) quantiles [e.g. the depth of the 10% quantile ( $q_{10}$ ) represents the shallowest depth above which 10% of the pollock  $s_A$  in the EDSU are encountered] differed more between vessels than the depths for higher (deeper) quantiles, with pollock





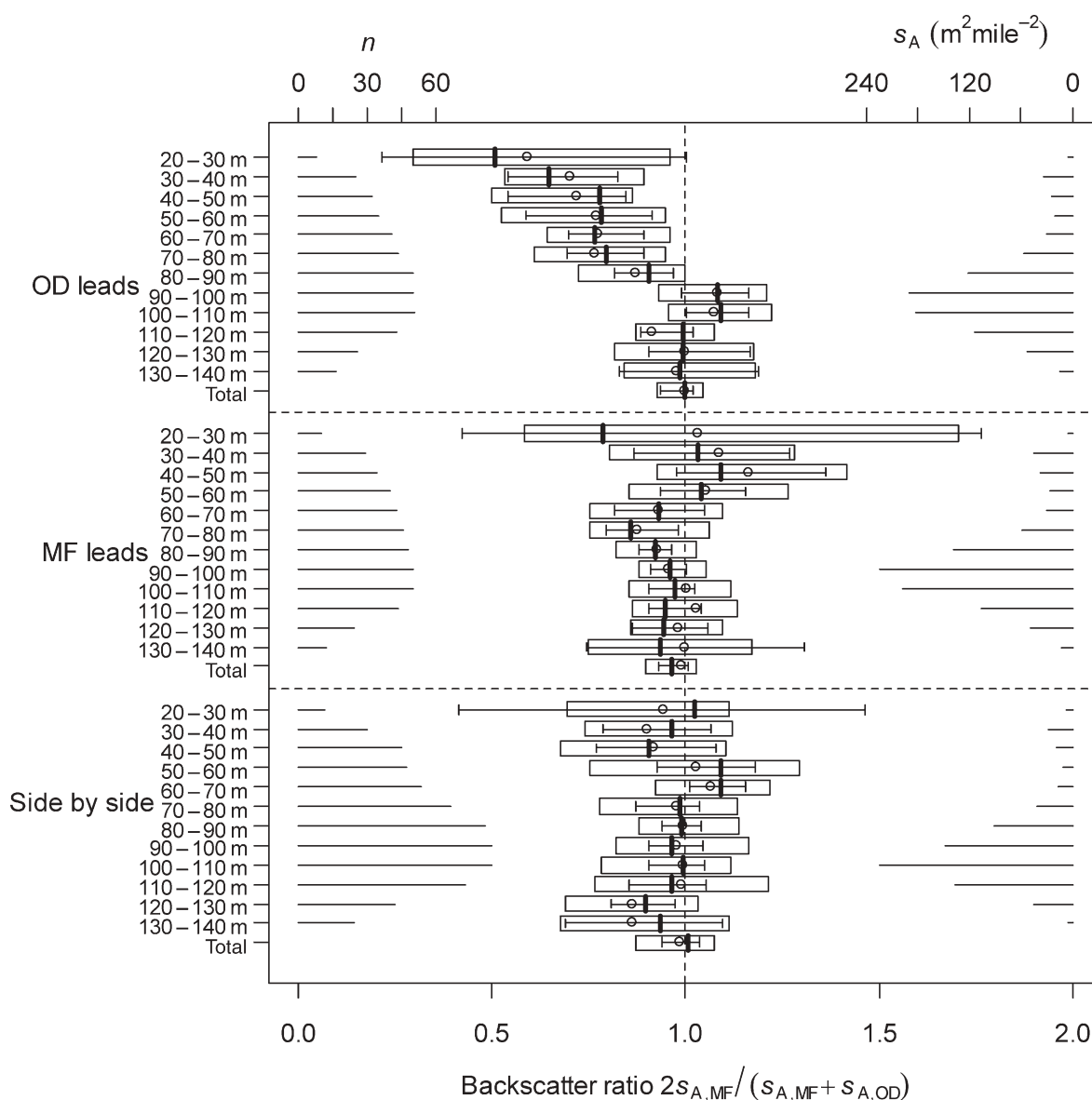
**Figure 6.** Box plots of vessel differences in quantile depths and mean depth of all observed pollock backscatter at 38 kHz for the follow-the-leader experiments when “Oscar Dyson” led (top) and “Miller Freeman” led (middle), and for the side-by-side transects (bottom). Each box shows the median (thick bar) with 95% CIs, the mean (circle), and the first and third quartiles (ends of box). For a given EDSU, the  $q\%$  quantile is the shallowest depth above which  $q\%$  of the pollock  $s_A$  is encountered. For example, the first bar in the uppermost plot indicates that the shallowest 10% of pollock backscatter in an EDSU five miles long was on average  $\sim 7$  m deeper when observed by MF than by OD when OD led a follow-the leader transect.

observed by the trailing MF being consistently deeper (Figure 6). This analysis suggests that the average vertical displacements were on the order of seven metres for the shallowest fish encountered in each EDSU five miles long (Figure 6), with the effect decreasing for fish deeper in the water column. The relatively larger increase in depth distribution for fish in shallower water indicates that pollock may have been diving in response to a stimulus, with pollock in shallower water responding more than those in deeper water. The diving response was supported by an analysis of pollock  $s_A$  in 10 m vertical strata that suggests that when OD led, MF detected fewer pollock at depths  $< 90$  m (Figure 7). However, there was no change in the total amount of pollock  $s_A$  observed over the water column. This suggests a vertical redistribution of fish but no change in echo intensity: the pollock in the upper  $\sim 90$  m had moved down by the time the following MF detected

them five minutes later. No strong patterns in vertical distribution were observed for the side-by-side measurements, or when the MF was in the lead (Figure 7).

## Discussion

The inter-vessel comparison of pollock and near-surface backscatter measured by the NOAA RVs “Miller Freeman” and “Oscar Dyson” revealed little overall difference in the abundance of pollock and near-surface backscatter attributable to vessel avoidance at the time of vessel passage. There were also no strong differences in the backscatter observed by the two vessels that could be related to latitude, longitude, mean fish depth, pollock abundance, or time of day. The measurements were made over an extensive area and over a broad range of conditions, representative of most situations encountered during “summer” pollock surveys



**Figure 7.** Box plots of the ratio  $2s_{A,MF}/(s_{A,MF} + s_{A,OD})$  recorded at 38 kHz in 10-m depth layers. Results are shown for pollock for the follow-the-leader experiments when “Oscar Dsyon” led (top) and when “Miller Freeman” led (middle), and for the side-by-side transects (bottom). Backscatter has been scaled such that the vessel ratios for the bottom echoes from both vessels are 1. Each box shows the median (thick bar) with 95% CIs, the mean (circle), and the first and third quartiles (ends of box). For each depth layer, only EDSUs where  $s_{A,MF} > 1$  and  $s_{A,OD} > 1$  were used. The number of such EDSUs is indicated at the left side of the plot. The vertical distribution of  $s_A$  for all EDSUs used to compute the vessel ratios in Figure 5 (including layer  $s_A < 1$ ) averaged over both vessels is given on the right. The ratio  $2s_{A,MF}/(s_{A,MF} + s_{A,OD})$  is plotted rather than the simpler  $(s_{A,MF}/s_{A,OD})$ , because this ratio is more stable when  $s_A$  is low. A value  $< 1$  indicates that MF detected fewer pollock than OD.

in the eastern Bering Sea. Our results indicate that for adult pollock surveyed under these conditions, it is unnecessary to correct survey-abundance estimates made with OD or MF for vessel-specific avoidance behaviour. However, this conclusion applies to adult pollock, because we did not encounter the high-density schools of juvenile pollock typically found in the area at shallower depths, depending on the strength of the year class (Honkalehto *et al.*, 2002).

Although we did not identify differences in pollock or near-surface mix backscatter attributable to differential vessel avoidance, we did observe small but statistically significant differences in echo-integration results between the two vessels, particularly

at 38 kHz, the primary frequency used for pollock surveys. Because these differences were not consistent across frequencies and were also evident in the bottom integrations, we conclude that they cannot be attributed to differential vessel avoidance, but more probably to instrument performance or calibration. Given the high precision in our on-axis calibrations, we suspect that the on-axis sphere calibration did not capture the full extent of uncertainty introduced by the instrumentation. Because we compared vessels, biases that affect results from both equally will have little impact on our results. One factor that may account for the discrepancy between the on-axis calibrations and the bottom-integration results is the equivalent beam angle

(EBA), which was based on manufacturer's measurements, as is common practice in fisheries acoustics (Simmonds and MacLennan, 2005). However, EBA can be affected by the transducer-mounting arrangement, but cannot be measured with high accuracy on vessel-mounted transducers (Simmonds, 1990; Reynisson, 1998). The on-axis method does not estimate EBA, and echo integration measurements such as bottom integration will reflect inaccuracies in EBA (Johannesson and Mitson, 1983). The uncertainty in the bottom-ratio adjustment and sphere calibration is not incorporated in the CIs given for the vessel ratios but, had it been accounted for, the CIs would have been slightly wider, which would reinforce our main conclusion of similar pollock backscatter from both vessels.

During the follow-the-leader experiments, we observed that, when OD led, pollock observed by MF were distributed deeper. This effect was more evident for shallower fish, which is consistent with a stronger response from fish closer to the surface, as is often observed during vessel avoidance (Vabø *et al.*, 2002). To help interpret the observation that MF detected fish deeper when OD was leading, but not *vice versa*, we consider two simple cases that are consistent with this observation: the first is that MF affected pollock depth most, with reaction primarily before transducer passage, and the second is that OD affected pollock depth most with reaction primarily after transducer passage. Depth differences, if any, were minor when the vessels were side by side, suggesting that the reaction took place primarily after transducer passage. In addition, the pollock backscatter summed over the water column did not differ between vessels in both side-by-side and follow-the-leader transects, suggesting that the change in backscatter depth was not solely due to changes in target strength caused by fish close to the surface changing their orientation. Therefore, we infer that the most likely scenario to explain the follow-the-leader observations is that pollock dived in response to OD, the reaction typically after transducer passage. The perturbation in the depth distribution would have to persist long enough for MF to detect the change in vertical distribution  $\sim 5$  min after the passage of OD and, owing to the delay in measurement, perhaps the response was in fact greater in magnitude than observed by MF. The observed disturbance by OD would not result in differences in echo-integration results during a standard survey under the experimental conditions, because the disturbance takes place after the transducer has passed the fish. However, perhaps reactions after vessel passage may alter the way fish interact with trawl gear towed behind a vessel (Wardle, 1993), potentially changing the size and composition of the catch (Fréon and Misund, 1999). To confirm the inference that pollock respond differentially to the vessels after passage, observations with a stationary echosounder are required (Godø and Totland, 1996; Ona *et al.*, 2007).

Our interpretation of the echo-integration comparisons depends on the assumption that the vessels were spaced far enough apart for the fish not to respond to a stimulus propagating from the accompanying vessel. For example, if during side-by-side measurements, the pollock observed by OD were actually responding to MF or *vice versa*, this would call both the vessel ratio and the depth distributions into question. Therefore, the validity of this assumption is a critical point, but the fact that the results were similar (i.e. similar vessel ratios) during the side-by-side and follow-the-leader experiments, where the separation distances increased from 0.5 to 1 mile, is reassuring, because one would expect vessel ratios to change with vessel spacing if one vessel influenced fish near the other. Additionally, if the stimulus is radiated

noise, propagation losses should make MF inaudible near OD, and *vice versa*, at these separation distances. Finally, observations of pollock avoidance to MF from a buoy-mounted echosounder suggest that avoidance behaviour is variable: on occasions where there was avoidance, subadult pollock did not appear to respond to MF at distances  $> 500$  m (C. Wilson, unpublished data). The closest vessel-separation distance used in this study was about twice that distance, suggesting that the fish observed by OD were unlikely to be reacting to the presence of the MF.

OD emits lower radiated noise than MF over a broad frequency range, but the follow-the-leader transects suggest that OD creates a larger response than MF after transducer passage. Although these results are unexpected, they are similar to the results of another inter-vessel comparison experiment, where a larger, noise-reduced vessel observed similar water-column backscatter from herring to that by a conventional vessel at the time of vessel passage, but caused a stronger diving response than the louder conventional vessel (Ona *et al.*, 2007). In that case, the largest differential reaction attributed to diving was also most evident after vessel passage (Ona *et al.*, 2007). Taken together with our observations, those results underscore the fact that the stimuli eliciting vessel avoidance behaviour remain poorly understood. The stimulus for the depth difference observed in our vessel comparison is unlikely to be radiated noise as defined under the ICES recommendation (Mitson, 1995), because OD produces much lower radiated noise than MF over a broad frequency range. Perhaps the stimulus is primarily auditory, but triggered by other characteristics of the sound, such as the rate of change of the sound, specific narrow-band tones, particle acceleration, or low-frequency infrasound (Enger *et al.*, 1993; Fréon and Misund, 1999). Fish have well-developed vision, and visual cues are a possibility. For example, vessel lights are known to alter fish availability to acoustic-survey methods at night (Lévénez *et al.*, 1990), but this is unlikely to explain our results, which span day and night. The pollock may be responding to visual stimuli from the vessel itself, but it is hard to understand how a visual stimulus could cause the observations, because contrast attenuation is likely to limit image-forming vision, but not light detection, to ranges of 10s of metres (Douglas and Hawryshyn, 1990). Although the vessels are of similar length, the OD has  $\sim 30\%$  more displacement than the MF, so more power is required to displace more water as the vessel moves, which may result in a greater disturbance for the fish to detect. Although the mechanism for such a potential disturbance and its detection by pollock remains unknown, our observations along with those of Ona *et al.* (2007) suggest that a mechanism other than the one-third-octave-band radiated noise levels underlying the ICES recommendation (Mitson, 1995) influences fish avoidance of survey vessels.

Although several noise-reduced vessels are now in operation (Mitson and Knudsen, 2003) and more are planned or under construction, the effectiveness of the measures taken in reducing vessel avoidance by fish remains poorly characterized. Our results indicate that a carefully designed and executed vessel inter-comparison is likely to be informative. Vessel inter-comparison is by no means a new approach (Johannesson and Mitson, 1983; Foote *et al.*, 1987), but we have adjusted typical protocols by spacing vessels such that the louder vessel is unlikely to influence the radiated noise field near the noise-reduced vessel, and by developing a combined side-by-side and follow-the-leader approach. This two-part approach allowed us to estimate effects during a survey using the side-by-side data, which reduced concern about the impact of

which vessel was leading, because this has complicated the interpretation of previous vessel-comparison studies (Wilson *et al.*, 2000). The follow-the-leader approach was originally developed to maximize spatial overlap between areas sampled by the vessels, but it also turned out to be useful to identify and interpret the effects after transducer passage.

The vessel-comparison approach has the disadvantage that the comparisons are relative, and cannot be referenced to an absolute standard. We can only establish if there are differences in echo-integration results between vessels, but we cannot distinguish, for example, whether pollock avoid both vessels to the same extent before transducer passage, or if pollock are not disturbed by a vessel passage at all. Another disadvantage of this comparative approach is that results are specific to differences between the vessels tested. For example, we cannot infer the extent of avoidance relative to undisturbed fish, or relative to studies conducted with other vessels. If such information is required, other approaches using instruments capable of detecting undisturbed fish are necessary, e.g. observations using sonars or stationary echosounders (Fréon and Misund, 1999).

Despite the limitations, the comparative approach is an effective method of establishing whether switching to a noise-reduced vessel will introduce major changes in a survey-abundance time-series. We plan to conduct several additional inter-comparison studies under different conditions. Pollock are surveyed at different seasons, and locations, and the fish are of different size, physiological and reproductive state than those investigated here. By conducting these additional studies, we hope to establish the impact of changing vessels on pollock-survey results under these different conditions. Vessel comparison will be most effective when it is combined with careful calibration and understanding of the measurement process.

Overall, we observed little evidence for vessel avoidance that will impact biomass estimates from echo-integration surveys of adult pollock during conditions encountered on summer EBS surveys when OD is used instead of MF. Analysis of depth distributions from both vessels suggests that there may be a comparatively stronger diving response to OD, with the reaction mainly in the upper 80–90 m, primarily after vessel passage. Because the response to the vessel appears to take place primarily after the fish are detected in the acoustic beam, this reaction is unlikely to influence acoustic survey results in this case. This observation was not anticipated, because the noise-reduced vessel appears to be associated with increased vessel avoidance. The increased response to a vessel designed to be stealthy highlights the need for improved understanding of the causes of vessel avoidance, and how reactions to vessels can be minimized. Progress on this front is likely to be made by reconsidering the potential stimuli produced by vessels in the context of the sensory mechanisms and behavioural reactions of fish.

## Acknowledgements

This work is the product of the efforts of many people without whom the study would not have been possible. Members of the AFSC acoustics group, particularly S. Furnish, R. Towler, P. Ressler, D. McKelvey, K. Williams, and D. Twohig, made tangible contributions. We are also indebted to D. Demer, L. Andersen, and E. J. Simmonds for advice and discussions related to calibration and echosounder performance. We thank the officers and crew of the NOAA ships “Miller Freeman” and “Oscar Dyson” for their assistance and careful ship-handling. We also

express our appreciation to those who prepared the vessels for the work, including members of the Pacific Marine Center, the US Navy Carderock Division, and NOAA Fisheries’ Office of Science and Technology, particularly M. Bancroft and E. Bradley. The comments of M. Jech, P. Ressler, R. O’Driscoll, J. Horne, and M. Dorn improved the manuscript. The work was supported by the Alaska Fisheries Science Center, NOAA. V. Hjellvik’s participation was made possible by the Norwegian Research Council through the Strategic Institute Program for IMR (“Absolute abundance estimation of fish” 143249/140).

## References

- Aglen, A. 1994. Sources of error in acoustic estimation of fish abundance. *In* Marine Fish Behaviour in Capture and Abundance Estimation, pp. 107–133. Ed. by A. Ferno, and S. Olsen. Fishing News Books, Blackwell Science, Oxford. 221 pp.
- Bahtiarian, M. 2005. Underwater noise. Not just a military problem. *Marine Technology Reporter*, July/August 2005: 40–48.
- Bailey, K. M., Quinn, T. J., Bentzen, P., and Grant, W. M. 1999. Population structure and dynamics of walleye pollock, *Theragra chalcogramma*. *Advances in Marine Biology*, 37: 179–255.
- Dalen, J., and Løvik, A. 1981. The influence of wind-induced bubbles on echo integration surveys. *Journal of the Acoustical Society of America*, 69: 1653–1659.
- De Robertis, A., and Wilson, C. D. 2006. Walleye pollock respond to trawling vessels. *ICES Journal of Marine Science*, 63: 514–522.
- Demer, D. A. 2004. An estimate of error for the CCAMLR 2000 estimate of krill biomass. *Deep-Sea Research II*, 51: 1237–1251.
- Douglas, R. H., and Hawryshyn, C. W. 1990. Behavioural studies of fish vision: an analysis of visual capacities. *In* The Visual System of Fish, pp. 373–418. Ed. by R. H. Douglas, and M. B. A. Djamgoz. Chapman and Hall, London. 562 pp.
- Engås, A., Misund, O. A., Soldal, A. V., Horvei, B., and Solstad, A. 1995. Reactions of penned herring and cod to playback of original, frequency-filtered and time-smoothed vessel sound. *Fisheries Research*, 22: 243–254.
- Enger, P. S., Karlsen, H. E., Knudsen, F. R., and Sand, O. 1993. Detection and reaction of fish to infrasound. *ICES Marine Science Symposia*, 196: 108–112.
- Fernandes, P. G., Brierley, A. S., Simmonds, E. J., Millard, N. W., McPhail, S. D., Armstrong, F., Stevenson, P., *et al.* 2000. Fish do not avoid survey vessels. *Nature*, 404: 35–36.
- Foote, K. G. 1985. Rather-high-frequency sound scattering by swim-bladdered fish. *Journal of the Acoustical Society of America*, 78: 688–700.
- Foote, K. G., Knudsen, H. P., Vestnes, G., MacLennan, D. N., and Simmonds, E. J. 1987. Calibration of acoustic instruments for fish-density estimation: a practical guide. *ICES Cooperative Research Report*, 144. 57 pp.
- Fréon, P., and Misund, O. A. 1999. Dynamics of Pelagic-Fish Distribution and Behaviour: Effects on Fisheries and Stock Assessment. Fishing News Books, Blackwell Science, Oxford. 348 pp.
- Godø, O. R., and Totland, A. 1996. A stationary acoustic system for monitoring undisturbed and vessel-affected fish behaviour. *ICES Document CM 1996/B*: 12. 11 pp.
- Gonzalez, A., Kollars, B., Harris, B., and Kipple, B. 1999. *Miller Freeman* (R/V 233) post-conversion/diagnostic-acoustic trial results. Naval Surface Warfare Center Carderock Division Research and Development Report Trial, 1999/187. 164 pp.
- Harmina, C. H. 2007. Acoustic-trial, radiated-noise results for *Oscar Dyson* (R-224) Naval Surface Warfare Center Carderock Division Research and Development Report Trial, 2007/529. 22 pp.
- Hazen, E. L., and Horne, J. K. 2004. Comparing the modelled and measured target-strength variability of walleye pollock, *Theragra chalcogramma*. *ICES Journal of Marine Science*, 61: 363–377.



- Hjellvik, V., and De Robertis, A. 2007. Vessel comparison on the seabed echo: influence of vessel attitude. US Department of Commerce, NOAA Technical Memorandum NMFS-AFSC. 34 pp. Available at <http://www.afsc.noaa.gov/Publications/AFSC-TM/NOAA-TM-AFSC-171.pdf>.
- Honkalehto, T., Williamson, N. J., de Blois, S., and Patton, W. 2002. Echo-integration, trawl-survey results for walleye pollock (*Theragra chalcogramma*) on the Bering Sea shelf and slope during summer 1999 and 2000. US Department of Commerce, NOAA Technical Memorandum NMFS-AFSC-125-126. 66 pp.
- Johannesson, K. A., and Mitson, R. B. 1983. Fisheries acoustics. A practical manual for aquatic-biomass estimation. FAO Fisheries Technical Paper, 240. 249 pp.
- Kieser, R., Mulligan, T. J., Williamson, N. J., and Nelson, M. O. 1987. Intercalibration of two echo-integration systems based on acoustic-backscattering measurements. Canadian Journal of Fisheries and Aquatic Sciences, 44: 562–572.
- Lévêque, J. J., Gerlotto, F., and Petit, D. 1990. Reaction of tropical, coastal pelagic species to artificial lighting and implications for the assessment of abundance by echo integration. Rapports et Procès-Verbaux des Réunions du Conseil International pour l'Exploration de la Mer, 189: 128–134.
- Misund, O. A. 1997. Underwater acoustics in marine fisheries and fisheries research. Reviews in Fish Biology and Fisheries, 7: 1–34.
- Mitson, R. B. (Ed.) 1995. Underwater noise of research vessels: review and recommendations. ICES Cooperative Research Report, 209. 61 pp.
- Mitson, R. B., and Knudsen, H. P. 2003. Causes and effects of underwater noise on fish-abundance estimation. Aquatic Living Resources, 16: 255–263.
- Olsen, K. 1990. Fish behaviour and acoustic sampling. Rapports et Procès-Verbaux des Réunions du Conseil International pour l'Exploration de la Mer, 189: 147–158.
- Ona, E., Godø, O. R., Handegard, N. O., Hjellvik, V., Patel, R., and Pedersen, G. 2007. Silent research vessels are not quiet. Journal of the Acoustical Society of America, 121: 145–150.
- Ona, E., and Mitson, R. B. 1996. Acoustic sampling and signal processing near the seabed: the deadzone revisited. ICES Journal of Marine Science, 53: 677–690.
- Ona, E., and Traynor, J. J. 1990. Hull-mounted, protruding transducer for improving echo integration in bad weather. ICES Document CM 1990/B: 31.
- Popper, A. N. 2003. Effect of anthropogenic sounds on fishes. Fisheries, 28: 24–31.
- Reynisson, P. 1998. Monitoring of equivalent beam angles of hull-mounted, acoustic-survey transducers in the period 1983–1995. ICES Journal of Marine Science, 55: 1125–1132.
- Simmonds, E. J. 1990. Very accurate calibration of a vertical echosounder: a five-year assessment of performance and accuracy. Rapports et Procès-Verbaux des Réunions du Conseil International pour l'Exploration de la Mer, 189: 183–191.
- Simmonds, E. J., and MacLennan, D. N. 2005. Fisheries Acoustics, 2nd edn. Blackwell Science, Oxford. 437 pp.
- Simrad. 2002. Non-linear effects: recommendation for fishery-research investigations. Simrad News Bulletin, March 2002.
- Simrad. 2003. Simrad ER60 Scientific echosounder application. Simrad AS, Horten, Norway. 166 pp.
- Skaret, G., Axelsen, B. E., Nøttestad, L., Fernö, A., and Johannessen, A. 2005. The behaviour of spawning herring in relation to a survey vessel. ICES Journal of Marine Science, 62: 1061–1064.
- Thomas, P. C., and Bradley, E. L. 2005. Acoustical trial results for *Oscar Dyson* (FSV 40) Naval Surface Warfare Center Carderock Division Research and Development Report Trial, 2005/549. 56 pp.
- Tichy, F. E., Solli, H., and Kaveness, H. 2003. Non-linear effects in a 200-kHz sound beam and the consequences for target-strength measurement. ICES Journal of Marine Science, 60: 571–574.
- Vabø, R., Olsen, K., and Huse, I. 2002. The effect of vessel avoidance of wintering Norwegian, spring-spawning herring. Fisheries Research, 58: 59–77.
- Wardle, C. S. 1993. Fish behaviour and fishing gear. In Behaviour of Teleost Fishes, pp. 609–643. Ed. by T. J. Pitcher. Chapman and Hall, New York. 740 pp.
- Wespestad, V. G., and Megrey, B. A. 1990. Assessment of walleye-pollock stocks in the eastern North Pacific Ocean: an integrated analysis using research-survey and commercial-fisheries data. Rapports et Procès-Verbaux des Réunions du Conseil International pour l'Exploration de la Mer, 189: 33–49.
- Wilson, C. D., Guttormsen, M. A., Cooke, K., Saunders, M. W., and Kieser, R. 2000. Echo-integration trawl survey of Pacific hake, *Merluccius productus*, off the Pacific coast of the United States and Canada during July–August, 1988. US Department of Commerce, NOAA Technical Memorandum NMFS-AFSC-118. 103 pp.

doi:10.1093/icesjms/fsn025